

Modules (Firebase Backend Architecture).

1. User & Host Module (Firebase Authentication & Profile)

- !3 User Registration and Secure Login via Firebase Authentication SDK
- !3 Firebase Email Verification Protection on Signup and Login Enforcement
- !3 Firebase User Token Authentication and Account Management
- !3 Deploy AI Assistant Bot to Zoom, Google Meet & Microsoft Teams
- !3 View Live Session Status & Audio Capture Stepper
- !3 View Speaker-by-Speaker Decrypted Transcript Timeline
- !3 Rename and Assign Custom Speaker Names across Transcript
- !3 View Executive Meeting Minutes (Agenda, Discussion Points, Decisions, Risks, Actions, Next Meeting)
- !3 Export Action Items to Spreadsheet CSV, Jira & GitHub Project Boards
- !3 Permanently Delete Meeting Recordings and Vault Data on Demand

2. Firebase Cloud Firestore Database & Security Vault Module

- !3 Firebase Cloud Firestore Document Database for Meetings, User Profiles, and Action Items
- !3 Firebase Security Rules and Auth Token Verification
- !3 Zero-Knowledge Client-Side Data Encryption before Cloud Storage
- !3 Real-Time Audio Stream Vault Protection
- !3 Data Integrity Verification Hash and Cryptographic Audit Log
- !3 Duplicate Meeting Detection and Warning Alerts before Processing

3. Speech Processing & Speaker Recognition Module

- !3 Accurate Speech-to-Text Transcription with Word Timestamps
- !3 Automatic Multi-Speaker Voice Separation and Diarization
- !3 Global Language Auto-Detection for 14+ Spoken Languages
- !3 Firebase Speaker Profile Registry and Recurring Participant Voice Auto-Recognition

4. Executive Minutes & Task Manager Module

- !3 Automatic AI Extraction of Structured Executive Briefings
- !3 Key Discussion Points, Agreed Decisions, and Identified Risks Summarization
- !3 Interactive Task Manager Board with Progress Completion Percentage Bar
- !3 Task Priority Classifications and Completion Status Management (High, Medium, Low)

5. Calendar Integration & Automation Module

- !3 Sync Upcoming Scheduled Meetings from Google Calendar & Microsoft Outlook
- !3 Automatic Meeting Link Extraction for Google Meet, Teams, and Zoom Room Links
- !3 Pre-Scheduled Bot Auto-Join 2 Minutes Prior to Meeting Start Time
- !3 Automated Email Dispatch of Executive Summary Reports to Meeting Participants

6. Natural Language Search & Meeting Analytics Module

- !³ Natural Language Semantic Search across Firebase Historical Meeting Archives
- !³ Relevance Percentage Match Scoring for Search Queries
- !³ Speaker Talk-Time Distribution and Speaking Share Charts
- !³ Speaking Speed Metrics (WPM), Inter-Speaker Pause Gaps, and Sentiment Insights

7. User Interface & Interactive Dashboards Module

- !³ Cybernetic Security Dashboard Interface
- !³ Interactive Spatial Node Map View (Neural Topology Canvas)
- !³ Multi-View Workspace Navigation (New Session, Previous Work, Spatial Topology, Cyber View)